

Assignment 4: Data Wrangling

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OVERVIEW

In this assignment, you will build a reproducible workflow that cleans four EPA air-quality files, combines them, reshapes the result, and produces grouped summaries. Date conversion is a brief review from Module 3; the emphasis is on data wrangling.

Directions

1. Rename this file `<FirstLast>_A04_DataWrangling.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. Ensure that code in code chunks does not extend off the page in the PDF.

Set up your session

Questions 1–2

- 1a. Load the `tidyverse` and `here` packages into your session.
 - 1b. Check your working directory.
 - 1c. Read all four raw files associated with the EPA Air dataset. See the README file for information about the variables and measurements.
2. Add the appropriate code to reveal the dimensions of the four datasets.

```
#1a
library(tidyverse)
library(here)
```

```
#1b
here()
```

```
## [1] "/home/guest/EDE_R_Lab_F26"
```

```
#1c
```

```
folder <- here('Data','Raw')
epafiles <- dir(folder, pattern = "EPAair*.*", full.names = TRUE)
EPADData <- epafiles %>%
  map(read.csv)
```

```
#2
```

```
colnames(EPADData[[1]])
```

```
## [1] "Date"
## [2] "Source"
## [3] "Site.ID"
## [4] "POC"
## [5] "Daily.Max.8.hour.Ozone.Concentration"
## [6] "Units"
## [7] "Daily.AQI.Value"
## [8] "Local.Site.Name"
## [9] "Daily.Obs.Count"
## [10] "Percent.Complete"
## [11] "AQS.Parameter.Code"
## [12] "AQS.Parameter.Description"
## [13] "Method.Code"
## [14] "CBSA.Code"
## [15] "CBSA.Name"
## [16] "State.FIPS.Code"
## [17] "State"
## [18] "County.FIPS.Code"
## [19] "County"
## [20] "Site.Latitude"
## [21] "Site.Longitude"
```

```
colnames(EPADData[[2]])
```

```
## [1] "Date"
## [2] "Source"
## [3] "Site.ID"
## [4] "POC"
## [5] "Daily.Max.8.hour.Ozone.Concentration"
## [6] "Units"
## [7] "Daily.AQI.Value"
## [8] "Local.Site.Name"
## [9] "Daily.Obs.Count"
## [10] "Percent.Complete"
## [11] "AQS.Parameter.Code"
## [12] "AQS.Parameter.Description"
## [13] "Method.Code"
## [14] "CBSA.Code"
## [15] "CBSA.Name"
## [16] "State.FIPS.Code"
```

```
## [17] "State"
## [18] "County.FIPS.Code"
## [19] "County"
## [20] "Site.Latitude"
## [21] "Site.Longitude"
```

```
colnames(EPADData[[3]])
```

```
## [1] "Date" "Source"
## [3] "Site.ID" "POC"
## [5] "Daily.Mean.PM2.5.Concentration" "Units"
## [7] "Daily.AQI.Value" "Local.Site.Name"
## [9] "Daily.Obs.Count" "Percent.Complete"
## [11] "AQS.Parameter.Code" "AQS.Parameter.Description"
## [13] "Method.Code" "Method.Description"
## [15] "CBSA.Code" "CBSA.Name"
## [17] "State.FIPS.Code" "State"
## [19] "County.FIPS.Code" "County"
## [21] "Site.Latitude" "Site.Longitude"
```

```
colnames(EPADData[[4]])
```

```
## [1] "Date" "Source"
## [3] "Site.ID" "POC"
## [5] "Daily.Mean.PM2.5.Concentration" "Units"
## [7] "Daily.AQI.Value" "Local.Site.Name"
## [9] "Daily.Obs.Count" "Percent.Complete"
## [11] "AQS.Parameter.Code" "AQS.Parameter.Description"
## [13] "Method.Code" "Method.Description"
## [15] "CBSA.Code" "CBSA.Name"
## [17] "State.FIPS.Code" "State"
## [19] "County.FIPS.Code" "County"
## [21] "Site.Latitude" "Site.Longitude"
```

```
dim(EPADData[[1]])
```

```
## [1] 9519 21
```

```
dim(EPADData[[2]])
```

```
## [1] 9466 21
```

```
dim(EPADData[[3]])
```

```
## [1] 9752 22
```

```
dim(EPADData[[4]])
```

```
## [1] 10184 22
```

```
glimpse(EPADData[[1]])
```

```
## Rows: 9,519
## Columns: 21
## $ Date                <chr> "03/01/2024", "03/02/2024", "03/0~
## $ Source              <chr> "AQS", "AQS", "AQS", "AQS", "AQS"~
## $ Site.ID             <int> 370030005, 370030005, 370030005, ~
## $ POC                 <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Daily.Max.8.hour.Ozone.Concentration <dbl> 0.028, 0.037, 0.038, 0.040, 0.045~
## $ Units               <chr> "ppm", "ppm", "ppm", "ppm", "ppm"~
## $ Daily.AQI.Value     <int> 26, 34, 35, 37, 42, 32, 38, 28, 2~
## $ Local.Site.Name     <chr> "Taylorsville Liledoun", "Taylors~
## $ Daily.Obs.Count     <int> 17, 17, 17, 17, 17, 17, 17, 17, 1~
## $ Percent.Complete   <dbl> 100, 100, 100, 100, 100, 100, 100~
## $ AQS.Parameter.Code  <int> 44201, 44201, 44201, 44201, 44201~
## $ AQS.Parameter.Description <chr> "Ozone", "Ozone", "Ozone", "Ozone~
## $ Method.Code        <int> 47, 47, 47, 47, 47, 47, 47, 47, 4~
## $ CBSA.Code          <int> 25860, 25860, 25860, 25860, 25860~
## $ CBSA.Name          <chr> "Hickory-Lenoir-Morganton, NC", "~
## $ State.FIPS.Code    <int> 37, 37, 37, 37, 37, 37, 37, 37, 3~
## $ State              <chr> "North Carolina", "North Carolina~
## $ County.FIPS.Code   <int> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, ~
## $ County             <chr> "Alexander", "Alexander", "Alexan~
## $ Site.Latitude      <dbl> 35.9138, 35.9138, 35.9138, 35.913~
## $ Site.Longitude     <dbl> -81.191, -81.191, -81.191, -81.19~
```

```
glimpse(EPADData[[2]])
```

```
## Rows: 9,466
## Columns: 21
## $ Date                <chr> "03/01/2025", "03/02/2025", "03/0~
## $ Source              <chr> "AQS", "AQS", "AQS", "AQS", "AQS"~
## $ Site.ID             <int> 370030005, 370030005, 370030005, ~
## $ POC                 <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Daily.Max.8.hour.Ozone.Concentration <dbl> 0.051, 0.044, 0.046, 0.045, 0.047~
## $ Units               <chr> "ppm", "ppm", "ppm", "ppm", "ppm"~
## $ Daily.AQI.Value     <int> 47, 41, 43, 42, 44, 36, 33, 42, 4~
## $ Local.Site.Name     <chr> "Taylorsville Liledoun", "Taylors~
## $ Daily.Obs.Count     <int> 17, 17, 17, 17, 17, 17, 17, 17, 1~
## $ Percent.Complete   <dbl> 100, 100, 100, 100, 100, 100, 100~
## $ AQS.Parameter.Code  <int> 44201, 44201, 44201, 44201, 44201~
## $ AQS.Parameter.Description <chr> "Ozone", "Ozone", "Ozone", "Ozone~
## $ Method.Code        <int> 47, 47, 47, 47, 47, 47, 47, 47, 4~
## $ CBSA.Code          <int> 25860, 25860, 25860, 25860, 25860~
## $ CBSA.Name          <chr> "Hickory-Lenoir-Morganton, NC", "~
## $ State.FIPS.Code    <int> 37, 37, 37, 37, 37, 37, 37, 37, 3~
## $ State              <chr> "North Carolina", "North Carolina~
## $ County.FIPS.Code   <int> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, ~
## $ County             <chr> "Alexander", "Alexander", "Alexan~
## $ Site.Latitude      <dbl> 35.9138, 35.9138, 35.9138, 35.913~
## $ Site.Longitude     <dbl> -81.191, -81.191, -81.191, -81.19~
```

```
glimpse(EPaData[[3]])
```

```
## Rows: 9,752
## Columns: 22
## $ Date          <chr> "01/01/2024", "01/04/2024", "01/07/2024~
## $ Source        <chr> "AQS", "AQS", "AQS", "AQS", "AQS", "AQS~
## $ Site.ID       <int> 370110002, 370110002, 370110002, 370110~
## $ POC           <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Daily.Mean.PM2.5.Concentration <dbl> 2.6, 2.4, 0.9, 1.1, 2.0, 3.0, 4.1, 2.8,~
## $ Units         <chr> "ug/m3 LC", "ug/m3 LC", "ug/m3 LC", "ug~
## $ Daily.AQI.Value <int> 14, 13, 5, 6, 11, 17, 23, 16, 5, 2, 12,~
## $ Local.Site.Name <chr> "Linville Falls", "Linville Falls", "Li~
## $ Daily.Obs.Count <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Percent.Complete <dbl> 100, 100, 100, 100, 100, 100, 100, 100,~
## $ AQS.Parameter.Code <int> 88502, 88502, 88502, 88502, 88502, 8850~
## $ AQS.Parameter.Description <chr> "Acceptable PM2.5 AQI & Speciation Mass~
## $ Method.Code     <int> 707, 707, 707, 707, 707, 707, 707, 707,~
## $ Method.Description <chr> "IMPROVE Module A with Cyclone Inlet-Te~
## $ CBSA.Code       <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ CBSA.Name       <chr> "", "", "", "", "", "", "", "", "", "",~
## $ State.FIPS.Code <int> 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,~
## $ State           <chr> "North Carolina", "North Carolina", "No~
## $ County.FIPS.Code <int> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11,~
## $ County          <chr> "Avery", "Avery", "Avery", "Avery", "Av~
## $ Site.Latitude   <dbl> 35.97235, 35.97235, 35.97235, 35.97235,~
## $ Site.Longitude  <dbl> -81.93307, -81.93307, -81.93307, -81.93~
```

```
glimpse(EPaData[[4]])
```

```
## Rows: 10,184
## Columns: 22
## $ Date          <chr> "01/01/2025", "01/04/2025", "01/07/2025~
## $ Source        <chr> "AQS", "AQS", "AQS", "AQS", "AQS", "AQS~
## $ Site.ID       <int> 370110002, 370110002, 370110002, 370110~
## $ POC           <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Daily.Mean.PM2.5.Concentration <dbl> 0.7, 1.2, 1.5, 4.2, 5.0, 8.5, 3.2, 3.2,~
## $ Units         <chr> "ug/m3 LC", "ug/m3 LC", "ug/m3 LC", "ug~
## $ Daily.AQI.Value <int> 4, 7, 8, 23, 28, 47, 18, 13, 41, 31~
## $ Local.Site.Name <chr> "Linville Falls", "Linville Falls", "Li~
## $ Daily.Obs.Count <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Percent.Complete <dbl> 100, 100, 100, 100, 100, 100, 100, 100,~
## $ AQS.Parameter.Code <int> 88502, 88502, 88502, 88502, 88502, 8850~
## $ AQS.Parameter.Description <chr> "Acceptable PM2.5 AQI & Speciation Mass~
## $ Method.Code     <int> 707, 707, 707, 707, 707, 707, 707, 707,~
## $ Method.Description <chr> "IMPROVE Module A with Cyclone Inlet-Te~
## $ CBSA.Code       <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ CBSA.Name       <chr> "", "", "", "", "", "", "", "", "", "",~
## $ State.FIPS.Code <int> 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,~
## $ State           <chr> "North Carolina", "North Carolina", "No~
## $ County.FIPS.Code <int> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11,~
## $ County          <chr> "Avery", "Avery", "Avery", "Avery", "Av~
## $ Site.Latitude   <dbl> 35.97235, 35.97235, 35.97235, 35.97235,~
## $ Site.Longitude  <dbl> -81.93307, -81.93307, -81.93307, -81.93~
```

All four datasets should have the same number of columns (PM2.5 has one extra column), but unique record counts (rows). Do your datasets follow this pattern? Yes.

Wrangle individual datasets to create processed files.

Questions 3–5

3. Change the Date columns to be date objects.
4. Select the following columns: DDate, Daily.AQI.Value, Local.Site.Name, AQS.Parameter.Description, County, Site.Latitude, Site.Longitude
5. For the PM2.5 datasets, fill all cells in AQS.Parameter.Description with “PM2.5” (all cells in this column should be identical).

```
#3
EPADData[[1]]$Date <- mdy(EPADData[[1]]$Date)
EPADData[[2]]$Date <- mdy(EPADData[[2]]$Date)
EPADData[[3]]$Date <- mdy(EPADData[[3]]$Date)
EPADData[[4]]$Date <- mdy(EPADData[[4]]$Date)
```

```
#4
glimpse(EPADData[[1]])
```

```
## Rows: 9,519
## Columns: 21
## $ Date                <date> 2024-03-01, 2024-03-02, 2024-03--
## $ Source              <chr> "AQS", "AQS", "AQS", "AQS", "AQS"~
## $ Site.ID             <int> 370030005, 370030005, 370030005, ~
## $ POC                 <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ Daily.Max.8.hour.Ozone.Concentration <dbl> 0.028, 0.037, 0.038, 0.040, 0.045~
## $ Units               <chr> "ppm", "ppm", "ppm", "ppm", "ppm"~
## $ Daily.AQI.Value     <int> 26, 34, 35, 37, 42, 32, 38, 28, 2~
## $ Local.Site.Name     <chr> "Taylorsville Liledoun", "Taylors~
## $ Daily.Obs.Count     <int> 17, 17, 17, 17, 17, 17, 17, 17, 1~
## $ Percent.Complete    <dbl> 100, 100, 100, 100, 100, 100, 100~
## $ AQS.Parameter.Code  <int> 44201, 44201, 44201, 44201, 44201~
## $ AQS.Parameter.Description <chr> "Ozone", "Ozone", "Ozone", "Ozone~
## $ Method.Code        <int> 47, 47, 47, 47, 47, 47, 47, 47, 4~
## $ CBSA.Code          <int> 25860, 25860, 25860, 25860, 25860~
## $ CBSA.Name          <chr> "Hickory-Lenoir-Morganton, NC", "~
## $ State.FIPS.Code    <int> 37, 37, 37, 37, 37, 37, 37, 37, 3~
## $ State              <chr> "North Carolina", "North Carolina~
## $ County.FIPS.Code   <int> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, ~
## $ County            <chr> "Alexander", "Alexander", "Alexan~
## $ Site.Latitude      <dbl> 35.9138, 35.9138, 35.9138, 35.913~
## $ Site.Longitude     <dbl> -81.191, -81.191, -81.191, -81.19~
```

```
dataset1 <-EPADData[[1]] %>%
  select(Date, Daily.AQI.Value, Local.Site.Name, AQS.Parameter.Description, County, Site.Latitude, Site
dataset2 <-EPADData[[2]] %>%
  select(Date, Daily.AQI.Value, Local.Site.Name, AQS.Parameter.Description, County, Site.Latitude, Site
dataset3 <-EPADData[[3]] %>%
```

```

select(Date, Daily.AQI.Value, Local.Site.Name, AQS.Parameter.Description, County, Site.Latitude, Site
dataset4 <- EPADData[[4]] %>%
  select(Date, Daily.AQI.Value, Local.Site.Name, AQS.Parameter.Description, County, Site.Latitude, Site

#5
dataset3$AQS.Parameter.Description <- "PM2.5"
dataset4$AQS.Parameter.Description <- "PM2.5"

```

Question 6

6. Save all four processed datasets in the Processed folder. Use the same file names as the raw files but replace “raw” with “processed”.

```

#6

write.csv(
  dataset1,
  row.names = FALSE,
  file = here("Data/Processed/dataset1.csv")
)

write.csv(
  dataset2,
  row.names = FALSE,
  file = here("Data/Processed/dataset2.csv")
)

write.csv(
  dataset3,
  row.names = FALSE,
  file = here("Data/Processed/dataset3.csv")
)

write.csv(
  dataset4,
  row.names = FALSE,
  file = here("Data/Processed/dataset4.csv")
)

```

Combine datasets

Question 7

7. Stack the four datasets with `bind_rows()`. Check that their column names are identical first.

```

#7

folder <- here('Data','Processed')
epafiles <- dir(folder, pattern = "dataset*.*", full.names = TRUE)
epacombined <- epafiles %>%
  map(read.csv) %>% ## read two csvs
  bind_rows() #binds csv's

```

Question 8

8. Use one pipeline (%>%) to create a daily site-level dataset that meets the following conditions:

- Include only sites that the four data frames have in common:

“Linville Falls”, “Durham Armory”, “Leggett”, “Hattie Avenue”,
“Clemmons Middle”, “Mendenhall School”, “Frying Pan Mountain”, “West Johnston Co.”, “Garinger High School”, “Castle Hayne”, “Pitt Agri. Center”, “Bryson City”, “Millbrook School”

(the function `intersect` can figure out common factor levels - but it will include sites with missing site information, which you don’t want...)

- Some sites have multiple measurements per day. Group by date, site name, AQS parameter, and county, then calculate the mean AQI value, latitude, and longitude. Handle missing values explicitly in each mean.
- Add numeric `Month` and `Year` columns from `Date`. Check if results are integers.
- Hint: the dimensions of this dataset should be $15,012 \times 9$.

#8

```
library(dplyr)
```

```
glimpse(epacombined)
```

```
## Rows: 38,921
## Columns: 7
## $ Date                <chr> "2024-03-01", "2024-03-02", "2024-03-03", "2~
## $ Daily.AQI.Value      <int> 26, 34, 35, 37, 42, 32, 38, 28, 25, 40, 44, ~
## $ Local.Site.Name      <chr> "Taylorsville Liledown", "Taylorsville Liled~
## $ AQS.Parameter.Description <chr> "Ozone", "Ozone", "Ozone", "Ozone", "Ozone",~
## $ County               <chr> "Alexander", "Alexander", "Alexander", "Alex~
## $ Site.Latitude        <dbl> 35.9138, 35.9138, 35.9138, 35.9138, 35.9138,~
## $ Site.Longitude       <dbl> -81.191, -81.191, -81.191, -81.191, -81.191,~
```

```
colSums(is.na(epacombined))
```

```
##           Date           Daily.AQI.Value           Local.Site.Name
##           0              0              0
## AQS.Parameter.Description           County           Site.Latitude
##           0              0              0
##           Site.Longitude
##           0
```

```
head(epacombined$Date, 20)
```

```
## [1] "2024-03-01" "2024-03-02" "2024-03-03" "2024-03-04" "2024-03-05"
## [6] "2024-03-06" "2024-03-07" "2024-03-08" "2024-03-09" "2024-03-10"
## [11] "2024-03-11" "2024-03-12" "2024-03-13" "2024-03-14" "2024-03-15"
## [16] "2024-03-16" "2024-03-17" "2024-03-18" "2024-03-19" "2024-03-20"
```

```

newdataset <- epacombined |>
  filter(Local.Site.Name %in% c(
    "Linville Falls",
    "Durham Armory",
    "Leggett",
    "Hattie Avenue",
    "Clemmons Middle",
    "Mendenhall School",
    "Frying Pan Mountain",
    "West Johnston Co.",
    "Garinger High School",
    "Castle Hayne",
    "Pitt Agri. Center",
    "Bryson City",
    "Millbrook School"
  )) |>
  filter(!is.na(Date)) |>
  mutate(
    Date = ymd(Date),
    month = month(Date),
    year = year(Date)
  ) |> group_by(Date, year, month, Local.Site.Name, AQS.Parameter.Description, County)|>
  summarise(meanAQI = mean(Daily.AQI.Value),
            meanlatitude = mean(Site.Latitude),
            meanlongitude = mean(Site.Longitude),
            .groups = 'drop')

```

Questions 9–10

9. Use `pivot_wider()` so that ozone and PM2.5 AQI values occupy separate columns. Each location-date combination should occupy one row.
10. Call up the dimensions of your new tidy dataset.

```

#9
newdatasetwider <- newdataset |>
  pivot_wider(
    names_from = AQS.Parameter.Description,
    values_from = meanAQI
  )
newdatasetwider

```

```

## # A tibble: 9,037 x 9
##   Date      year month Local.Site.Name      County meanlatitude meanlongitude
##   <date>    <dbl> <dbl> <chr>          <chr>      <dbl>      <dbl>
## 1 2024-01-01  2024     1 Bryson City    Swain        35.4      -83.4
## 2 2024-01-01  2024     1 Castle Hayne  New H~       34.4      -77.8
## 3 2024-01-01  2024     1 Clemmons Middle Forsy~       36.0      -80.3
## 4 2024-01-01  2024     1 Durham Armory Durham       36.0      -78.9
## 5 2024-01-01  2024     1 Frying Pan Mountain Haywo~       35.4      -82.8
## 6 2024-01-01  2024     1 Garinger High School Meckl~       35.2      -80.8
## 7 2024-01-01  2024     1 Hattie Avenue Forsy~       36.1      -80.2
## 8 2024-01-01  2024     1 Leggett      Edgec~       36.0      -77.6

```

```
## 9 2024-01-01 2024 1 Linville Falls Avery 36.0 -81.9
## 10 2024-01-01 2024 1 Mendenhall School Guilf~ 36.1 -79.8
## # i 9,027 more rows
## # i 2 more variables: PM2.5 <dbl>, Ozone <dbl>
```

```
#10
```

```
dim(newdatasetwider)
```

```
## [1] 9037 9
```

Question 11

11. Save your processed dataset with the following file name: “EPAair_O3_PM25_NC2425_Processed.csv”

```
#11
```

```
write.csv(newdatasetwider,
          row.names = FALSE,
          file = here("Data/Processed/EPAair_O3_PM25_NC2425_Processed.csv")
)
```

Generate summary tables

Questions 12–13

12. Create a monthly summary data frame grouped by site, month, and year. Calculate mean ozone and PM2.5 AQI for each group, handling missing values explicitly. Then use `drop_na()` to remove rows only when mean **ozone** is missing. Mean PM2.5 may remain missing.
13. Call up the dimensions of the summary dataset.

```
#12
```

```
monthlysummary <- newdatasetwider |>
  filter(!is.na(Ozone) & !is.na(PM2.5)) |>
  group_by(Local.Site.Name, month, year) |>
  summarise(
    meanozone= mean(Ozone),
    meanpm2 =mean(PM2.5),
    .groups = 'drop'
  )
```

```
#13
```

```
dim(monthlysummary)
```

```
## [1] 234 5
```

Question 14

14. Why is `drop_na(mean_ozone)` more appropriate here than `na.omit()`? Test both approaches and compare the dimensions of the resulting summary data frames.

Answer: `drop_na(mean_ozone)` is more appropriate because it removes only observations where mean ozone is missing, while `na.omit()` removes observations with missing values in any column. `na.omit()` gets rid of observations that have ozone data but are missing values for other variables.